

TABLE OF CONTENTS

M-432 OVERHEAD AND GANTRY CRANES

	Page No.
1 GENERAL	M 432-3
2 SCOPE OF WORK.....	M 432-3
3 CODES AND STANDARDS	M 432-4
4 DESIGN AND OPERATING CONDITIONS	M 432-4
4.1 Steam Turbine Overhead Travelling Bridge Crane	M 432-5
4.2 Cooling Water Pump Overhead Travelling Bridge Crane	M 432-6
4.3 Cooling Water Intake Gantry Crane.....	M 432-6
5 DETAILED REQUIREMENTS	M 432-7
5.1 Structural	M 432-7
5.2 Walkways, Handrails, Toe Plates and Fixed Ladder	M 432-7
5.3 Operator's Cab (For Overhead Crane)	M 432-8
5.4 Bridge End Truck and Bridge Drive.....	M 432-8
5.5 Runway Stops	M 432-9
5.6 Bridge Rails.....	M 432-9
5.7 Trolley	M 432-9
5.8 Wheels	M 432-9
5.9 Hoisting Equipment	M 432-9
5.10 Gearing	M 432-10
5.11 Brakes	M 432-10
5.11.1 Bridge Motion Brakes.....	M 432-10
5.11.2 Hoist Motion Brakes	M 432-10
5.12 Bearings.....	M 432-10
5.13 Lubrication	M 432-10
5.14 Guards.....	M 432-11
5.15 Miscellaneous	M 432-11
5.16 Painting.....	M 432-11
5.17 Protective Coating	M 432-11
6 PERFORMANCE TESTING AND GUARANTEES	M 432-11
6.1 Shop Test	M 432-11
6.2 Site Test.....	M 432-12
6.2.1 Running in Period Test	M 432-12
6.2.2 Functional Test at Site	M 432-12
6.2.3 Alignment, Breakage and Excessive Wear	M 432-12
7 QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)	M 432-12
7.1 General	M 432-13
7.2 Procedures and Documentation	M 432-13
7.3 Examinations.....	M 432-13

	7.3.1	Hooks	M 432-13
	7.3.2	Structural Work	M 432-13
8		INSTRUMENTATION AND CONTROLS	M 432-13
9		ELECTRICAL SPECIFICATIONS	M 432-14
	9.1	General	M 432-14
	9.2	Motors.....	M 432-15
	9.3	Power Panel.....	M 432-15
	9.4	Conduits, Cables and Wiring	M 432-15
	9.5	Limit Switches	M 432-15
	9.6	Lighting and Convenience Outlet Equipment.....	M 432-16
	9.7	Trolley and Bridge Conductors.....	M 432-16

M 432 OVERHEAD AND GANTRY CRANES

1 GENERAL

This section of the specification describes the technical requirements for Overhead and Gantry Cranes for The Muara Tawar combined cycle power plant (Block 2,3&4) Add-On Project, and covers the minimum requirements for the design, engineering, manufacture, erection and inspection and testing.

Overhead and Gantry Cranes shall be complete in every respect and suitable for safe reliable and continuous operation.

This specification does not enumerate or describe all the materials and equipment to be supplied and all the services to be performed. All materials and equipment shall be provided as are required to make a complete, properly functioning installation and shall conform to the highest standards of engineering design and workmanship.

2 SCOPE OF WORK

The Scope of Work shall be in accordance with Part 4, subsection 1.2. Summary of work, and sets out the scope of the equipment and services as well as specially requested supplies and services but it in no way excludes other necessary components and services not specifically mentioned herein.

The Bidder shall provide and supply the Overhead and Gantry Cranes complete, including all necessary components, spare parts and services.

The Scope of supply and services shall include the design, manufacture, construction, factory testing, packing for transport, transport and delivery to site, unloading at site, storing, complete erection, testing at site, painting, commissioning, putting into normal operation, tie-in with existing facilities, reliability and performance testing of cranes and hoists and associated equipment including but not limited to:

- One (1) Electric Overhead Traveling Bridge Cranes (EOT) for the Steam turbine building; one (1) for block 2, 3 & 4.
- One (1) Electric Overhead Traveling Bridge Cranes; for covering Block 2, 3 & 4 C.W. pump and C.W. pump motor.
- One (1) set extended rail of Gantry Cranes to carry, assemble or disassemble stop logs, bar screens; for covering Block 2, 3 & 4. The existing Gantry Cranes will be used for block 1, 2, 3 & 4.

Complete electric equipment as detailed in clause 9 herein.

3 CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.

The Codes and Standards from the Part 4, Subsection 2.5 “Codes and Standards” or other equivalent international standards are specifically applicable to the design, manufacture and testing of the Work included in this Specification.

AISE	Association of Iron and Steel Engineers
AISI	American Iron and Steel Institute
ASTM	American Society for testing and Material
NEC	National Electrical Code, USA
NEMA	National Electrical Manufacturers Association, USA
SSPC	Structural Steel Painting Council
IEC	International Electrotechnical Commission
CMAA Specification No. 70	Specifications for Electric Overhead Traveling Cranes
ANSI B30.2.0	Safety Standard for Overhead and Gantry Cranes (Top Running Bridge, Multiple Girder).
AWS D1 .1	Structural Welding Code
AISC	Specification for the Design, Fabrication and Erection of Structural Steel for Buildings
OSHA Standard	Part 1910 General Industry Standards
DIN	Deutsches Institut Für Normung

4 DESIGN AND OPERATING CONDITIONS

All structural and mechanical parts of the crane shall be designed to resist dead and live loads, impact allowance, and torsional lateral forces, and shall take into account the condition of the installation area.

The cranes shall be equipped with all necessary motors, gearing, and controlling devices for all movements of the bridge, trolley, hoists, control

station. The Turbine Building Cranes shall be equipped with operator's cab and the CW pump house crane shall be pendant controlled.

The overhead crane service classification shall be Class A (Standby Service) in accordance with CMAA Specification No. 70 or equivalent International Standard.

The cranes shall be especially suited for power plant service requiring precise control of all movements in handling and positioning of large power plant machinery, parts and materials during initial construction and subsequent disassembly and reassembly of the machinery during overhaul periods. Crane operation shall be smooth and without jerky movements. After the erection period of the power station, a complete inspection, repair (if necessary) and overhaul of the crane shall be carried out by the Bidder to the satisfaction of the Owner.

All parts shall be readily accessible for inspection, adjustment and repair.

The cranes shall be capable of raising, lowering and holding in position and transporting loads equal to the rated load over the entire rated load and speed range without damage or excessive deflection to any crane part.

The rated load of the cranes shall be marked on each side of the crane and shall be visible from the operating floor. Each hoist shall have its rated load marked on it or on its load block.

The cranes shall be designed and constructed suitable for the following conditions. The bidder shall verify these data and provide confirmed data with Bid Proposal.

4.1 Steam Turbine Overhead Travelling Bridge Crane

(1)	Installation	Indoor
(2)	Type	Overhead Traveling Bridge Crane
(3)	Number	One (1) set for Block 2, 3&4.
(4)	Capacity	Main Hoist 100 tons (Approx.) Auxiliary hoist 20 tons
(5)	Speed	Main hoist (max.) 1 .2 m/min Auxiliary hoist 5.0 m/min Trolley travel (max.) 20 m/min Bridge travel 20.0 m/min
(6)	Rail span	Bidder to provide .

(7)	Height from floor	Bidder to provide to top of rail
(8)	Length of rails	Bidder to provide
(9)	Vertical control	max.3 mm movement
(10)	Motor	Enclosed type for crane
(11)	Operation	From closed Cabin

4.2 Cooling Water Pump Overhead Travelling Bridge Crane

(1)	Installation	Indoor
(2)	Type	Overhead traveling crane
(3)	Number	One (1) set for covering Block 2, 3, 4
(4)	Capacity	Main Hoist 25 tons (Minimum) Auxiliary hoist 5 tons
(5)	Speed	Main hoist (max.) 5 m/min Auxiliary hoist 10 m/min Trolley travel (max.) 20 m/min Bridge travel 32.0 m/min
(6)	Rail span	Bidder to provide
(7)	Height from floor	Bidder to provide To top of rail
(8)	Length of rails	Bidder to provide
(9)	Vertical control movement	max. 3 mm
(10)	Motor	Enclosed type for crane use
(11)	Operation	Pendant push button

4.3 Cooling Water Intake Gantry Crane

(1)	Type	Gantry Type (existing)
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(2)	Number	One (1) set extended rail of Gantry Cranes	
(3)	Rail length	Bidder to provide	
(4)	Capacity	15 tons	ADD1 -IIC -004
The Bidder shall provide all miscellaneous crane and hoist as necessary.			

5 DETAILED REQUIREMENTS

5.1 Structural

All structural steel shall be ASTM A36.

Structural design shall conform to the requirements of CMAA Specification No.70 and the AISC Manual. In the event of a conflict between the two documents, the more conservative requirement shall govern the design.

All shop connections on the crane structure shall be made by welding, bolting or any suitable combination thereof. All field connections shall be bolted.

All welding shall be performed per AWS D1.1.

The welding procedures shall be in accordance with AWS D1.1 or Section IX of ASME.

High strength bolts shall be used for all structural connections which are not welded. Bolts shall be ASTM A325 or A490 or equal utilizing friction type connections.

All load bearing connections not utilizing high strength bolts shall use ASTM A307 bolts or equal.

Bidder shall furnish all bolts, nuts and washers which are required for field erection of the crane.

5.2 Walkways, Handrails, Toe Plates and Fixed Ladder

Overhead Cranes shall be provided with walkways complete with handrails, toe plates and fixed ladders where required for regular access to operating machinery for maintenance.

Material for walkways, handrails, toe plates and fixed ladders shall be ASTM A36 steel or equal. Walkways shall be made of chequered steel plate.

Gantry Crane shall be provided with a walkway for maintenance.

5.3 Operator's Cab (For Overhead Crane)

The crane shall be provided with an enclosed type cab. The cab shall contain all operating devices complete with static stepless controls for all movements of bridge, trolley and hoists.

The cab design shall also include the following :

- The frames and enclosures shall be “solidly grounded”.
- One package air conditioning, to be operated from a 220 V convenient outlet (AC single phase), and controlled by a switch in the cab.
- One chair, permanently mounted and fully adjustable, located to provide ease of operation of all the crane control devices which are located in the cab.
- A Rescumatic harness or equal acceptable to the Owner, to allow emergency egress from the cab to the floor from any crane position.
- The cab floor shall be made of steel plate, covered with electrically non-conductive, hard-wearing, asbestos-free material to Manufacturer's standard.
- A fixed access ladder shall be provided from the cab floor to the crane foot walk. The ladder shall be in compliance with Clause 5.2 contained herein.
- A main disconnect switch shall be provided in the cab for disconnecting all Power except lighting.
- An automatic/manual electric warning gong, 250 mm diameter minimum, mounted outside the cab. It shall be operated automatically and shall be used for warning purpose each time before the crane travels and intermittently during travel.

5.4 Bridge End Truck and Bridge Drive

The crane shall be designed so that either wheel of a truck may be removed without removing the adjacent truck.

The end trucks shall be designed for rotating axles only.

End trucks shall be equipped with rail sweeps extending below the rail head and designed to move an obstruction upward and away from the rails.

Jacking pads shall be provided for use in jacking the trucks for wheel removal.

5.5 Runway Stops

Runway stops shall be furnished and installed by the Bidder for mounting on the runway girders at the limits of the bridge travel.

5.6 Bridge Rails

The bridge rails shall be fastened in place with clips to maintain the center distance of rails.

5.7 Trolley

The trolley shall contain all mechanisms for operating the hooks and for moving the trolley on the bridge rails. The trolley frame shall be constructed of welded structural steel only. The use of cast iron or cast steel is not acceptable.

A suitable handrail shall be provided around top of trolley, except if trolley can be normally serviced from crane walkways or platforms in which case a handrail around top of trolley will not be required.

Trolley trucks shall be equipped with rail sweeps extending below the rail heads and designed to move an obstruction upward and away from the rails.

The design of the trolley deck shall provide accessibility to the sheaves for purposes of inspecting and maintaining the hoisting ropes.

5.8 Wheels

Wheels shall be of rolled or forged steel only. Cast iron or cast steel shall not be used.

Wheels shall be keyed to the axles. Bearings shall be anti-friction type provided with proper lubrication.

5.9 Hoisting Equipment

Drums and Sheaves: Double fall reeving shall be used for main hoist. Single fall reeving shall not be used.

The hoisting ropes shall be in accordance with CMAA Specification No. 70.

The hoisting ropes shall be fabricated from Improved Plow Steel.

Each hook shall be forged steel supported on a ball or roller bearing. The hook shall rotate freely on this bearing.

5.10 Gearing

All gears shall be steel with cut teeth and shall be of the helical or herringbone type.

5.11 Brakes

The crane shall be provided with brakes to retard or stop and hold the motion of the bridge, trolley and hoists. Brakes shall be suitably mounted for convenient and rapid operation.

5.11.1 Bridge Motion Brakes

The bridge shall be provided with one (1) electric/hydraulic shoe-type brake per motor with a torque rating of 150% of the bridge motor.

5.11.2 Hoist Motion Brakes

The Bidder shall furnish the following:

- Two electric shoe-type holding brakes for each hoisting unit, each with a torque rating of at least 100% of the motor. One brake shall be located on the motor shaft and the other shall be on the motor shaft extension on the opposite side of the gear case.
- The brake for the trolley motion may be either the electric shoe-type or the electric disc-type. The torque rating of the brake shall comply with CMAA Specification 70.

5.12 Bearings

All bearings, except those subject to a small amount of movement, shall be of the anti-friction ball or roller type.

Bearings which are associated with enclosed gears which either run in oil or are splash lubricated shall also be oil bath lubricated. Other bearings, such as axle bearings, cross-shaft bearings and sheave bearings, shall be packed with grease and lubricated through pressure-type fittings. Bearing enclosures shall be designed to exclude dirt and prevent leakage of oil or grease.

Anti-friction bearings shall be selected to give a minimum B-10 life of 50,000 hours and based on the full rated speed.

5.13 Lubrication

Lubricating devices shall be provided for all bearings and journals: Bearing housings shall be oil-and/or grease-tight. The equipment shall be arranged so that all points of lubrication are easily and safely accessible. Sheet steel catch pans and guards shall also be provided to prevent the dripping or splashing of oil and grease.

Lubrication points and gear box drains shall be piped to accessible locations.

Bidder shall furnish a sufficient quantity of all lubricants required for the equipment furnished by him for initial operation of the equipment.

5.14 Guards

Shields or enclosure shall be provided over any part of the equipment which needs protection from water and dust. All slip rings, limit switches and other contact-making devices shall be well protected.

The trolley frame shall be provided with a guard to prevent the hoist ropes from swinging into the bridge conductors.

The bridge shall be provided with a guard to prevent the hoist ropes from swinging into the runway conductors.

5.15 Miscellaneous

After the assembly test and before disassembling, the shop assembly of the crane shall be approved by the Owner and connecting parts shall be match-marked to facilitate field erection. Match-marks shall be shown on erection drawings furnished by the Bidder. Each part designated on the erection drawings by a mark number shall be clearly and legibly marked in a conspicuous place with the appropriate number if it is to be shipped separately.

5.16 Painting

All parts of the equipment and structures shall be thoroughly cleaned as required by Steel Structure Painting Council Procedure SSPC-SP10 for near white Metal Blast Cleaning of all mill scale, rust, grease and other foreign matter, and shall be given a shop and field coats of paint as indicated in Part 4, Specifications S-420 and S-430.

5.17 Protective Coating

Before shipment all exposed machined surfaces shall be coated with a suitable grease or anti rust compound.

Steel surfaces left unpainted in the shop shall be coated with a corrosive-resistant material.

6 PERFORMANCE TESTING AND GUARANTEES

6.1 Shop Test

The cranes shall be completely assembled in shop . All motors controls and machinery shall be connected and all machinery on the crane shall be operated under its own power at rated voltage. The volts, amperes and watts shall be recorded for each operation. Electrical circuit and interlocks shall be tested for

correct operation and sequence and all test shall be witnessed and approved by the Owner.

6.2 Site Test

6.2.1 Running in Period Test

The cranes shall be tested by the Bidder to the requirements of the Owner. Before starting the test the hoist shall be operated under light loads for at least two (2) hours in order to run-in the gearing and to ensure proper lubrication. After this run in period the cranes shall be inspected to determine if all parts have remained in correct alignment and no excessive wear has occurred.

6.2.2 Functional Test at Site

Volts, amperes and watts input shall be recorded for each applicable motor during all of the test and compared with the values obtained during the shop test. To avoid the reconnecting electrical meters for each motion of the crane, the meters may be connected at the incoming power supply of the crane. The amount of power used for crane accessories should be determined prior to testing and on completion of the tests.

The Bidder shall be responsible for obtaining suitable test weights for use in testing. Verification and recordings of these weights shall be in presence of the Owner. The following test shall be performed on the crane by the Bidder under the direction of the Owner.

The hoist shall be operated to raise, lower and hold in any position a test load of 125 % of the rated capacity.

The load shall be both raised and lowered at both creep and normal speeds of hoisting.

The cranes shall be operated to transport test loads of 100% of the rated capacity for the full available length of the runway.

Electrical input and hoisting and lowering speed data shall be compiled and recorded. All interlocks and safety devices shall be tested.

In addition a deflection check of each main girder frame shall be made with the cab in its central position supporting the maximum safe working load.

6.2.3 Alignment, Breakage and Excessive Wear

On completion of the tests on the crane the Bidder shall check the crane for alignment, breakage and excessive wear.

7 QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

7.1 General

Quality assurance/control shall conform to the requirements of Part 4, Subsection 2.7.

7.2 Procedures and Documentation

The quality assurance/control procedures and documentation specified shall include but not be limited to the following:

- Welding
- Non destructive testing
- Heat treating
- Packaging and shipping
- Shop Test procedures
- Field Test procedures
- Cleaning and Painting

7.3 Examinations

7.3.1 Hooks

In addition to manufacturer's normal quality control inspection procedure for ensuring high quality crane hooks, the main hook and auxiliary hook shall be given an ultrasonic examination in the finished form plus a magnetic particle and/or liquid penetrant examination.

7.3.2 Structural Work

All full penetration welds shall be 100% radiographed, in accordance with acceptable standards. Radiographs shall be submitted to the Owner for review and acceptance. All other welds shall be visually inspected.

8 INSTRUMENTATION AND CONTROLS

Control equipment of the five step magnetic or static stepless type shall be supplied and conveniently located on the bridge or trolley. Approved covers and housings shall be provided to protect all electrical equipment from dust and dripping water. All open type resistors shall be mounted in dripproof ventilated enclosures. All control equipment enclosures shall have thermostatically controlled, 220 V AC space heating equipment.

Reversing magnetic contactors, not smaller than NEMA Size 1, shall be provided for each motor. Contactor shall be provided with necessary auxiliary switches and overload protection.

The cranes operation will be from either an operator's cab, or a pendant push-button station. All master controls on the pendant push button station shall have "dead man" spring return feature.

Nameplates shall be provided for readily identifying the purposes of each piece of control equipment and to indicate the direction of movement which will result from operating push-buttons.

Pendant pushbutton stations shall be provided with a ground terminal connected to a separate conductor in the cable to the control equipment to ground the pushbutton enclosure.

Hoist Controls:

The main and auxiliary hoists shall be provided with static stepless control. A separate control shall be furnished for each motion. Minimum movements shall be 0.3 mm for the main hoist and 1.8 mm for the auxiliary hoist.

The crane shall be capable of raising, lowering and holding in position and transporting loads equal to rated load without damage or excessive deflection to any crane part.

Bidder shall provide necessary instrumentation to the cranes to enable them to operate under all conditions.

All electrical and control equipment shall meet the requirements of Technical Specifications E-001, I-001 and I-002

9 ELECTRICAL SPECIFICATIONS

9.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for The Overhead and Gantry cranes.

Those equipment shall include, but not be limited to the followings:

1. Motors
2. Power Panel
3. Conduits, cables and wiring
4. Limit Switches
5. Lighting and convenience socket outlet equipment
6. Trolley and Bridge Conductors
7. All electrical control, monitoring and instrumentation devices ,and other appurtenances required for this Specification, shall be provided by the Bidder.

For general electrical requirement, See Part 4, Technical Specification E-001.

9.2 Motors

The drive motors shall meet the following specific requirements as well as the general applicable requirements of Technical Specification E-210 Low Voltage Motors. In case of conflict, the following specific requirements will apply:

Rated horsepower of motors shall be at least 120% of the maximum brake horsepower required by the driven equipment. This relation shall exist for all operating speeds and conditions. The motors shall be rated for continuous duty (IEC duty type S1).

Motors shall be capable of developing a breakdown torque in accordance with NEMA MG1-18.509.

9.3 Power Panel

For detailed electrical requirements, see Part 4, Technical Specification E-001 and E-280.

9.4 Conduits, Cables and Wiring

All motors and electrical, control, monitoring and instrumentation devices furnished and installed shall be wired to the power and local control panel respectively.

The crane shall be shipped completely wired at the factory except for those connections that need to be disconnected for shipment. Disconnected wires shall be furnished with terminal tugs installed. All conduit and wiring connections shall be properly tagged for convenient field assembly during erection.

For detailed electrical requirement, see Part 4, Technical Specifications E-230 and E-232.

9.5 Limit Switches

The bridge and trolley shall be provided with spring action limit switches of the automatic reset type to prevent overtravel in either direction. The main hoist shall be provided with limit switches to stop the hook in its highest and lowest safe positions.

Each limit switch shall be so wired that the drive motor can be energized in the reverse direction after its limit switch is opened. Two (2) limit switches shall be used to limit the upward travel of the hoist hooks. Limit switches shall be subject to the Owner's review and acceptance.

9.6 Lighting and Convenience Outlet Equipment

Lights be provided on top of the bridge along the walkway together with underbridge floodlights to illuminate the area below the crane.

All fixtures shall be easily accessible for lamp replacement, and shall be suitably protected against excessive vibration. The underbridge floodlights shall have swing-up type brackets accessible from the bridge walkway. The lamp voltage for the cab, walkway and underbridge lights shall be 220 V AC. The underbridge floodlights shall be subject to Owner's review.

Enclosed circuit breakers of the automatic thermal- magnetic type shall be provided for each lighting circuit. The underbridge floodlights shall be on a separate circuit from all others.

Two(2) 250 V rated, 16A, 3-pin grounding type standard plug receptacles as per IEC 60309 shall be furnished; one located on the bridge end, if applicable, one located in the operator's cab. The receptacles shall be fed from a separate circuit breaker independent of the lighting circuits.

9.7 Trolley and Bridge Conductors

A trolley power system shall be furnished, consisting of solid copper conductors each enclosed in an insulated protective cover complete with supporting insulators and power line connector for incoming power cable. The trolley shall run the full length of the crane travel.

Bridge span conductors shall consist of a festooned system of rubber covered flexible copper cable, supported by a group of trolleys running on a beam track. The system shall be mounted on the outside of the crane girder. Each lead of the festoon cable shall be tagged and permanently lugged with compression lugs.

(Last page of M-432)